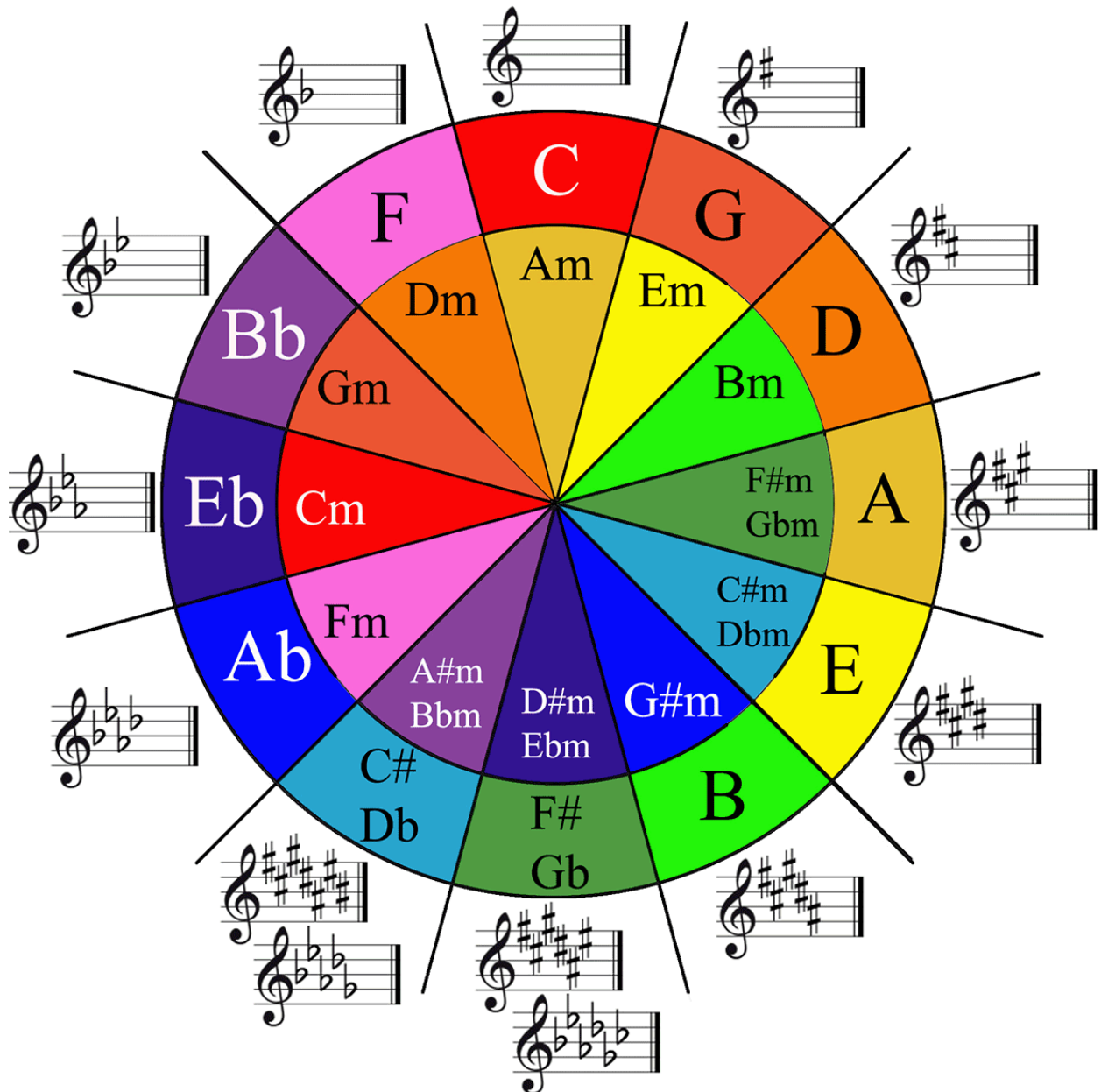


Emergent Tonal Structure from a Simple Neural Network

Colab: [CGOS 269 - Emergent Tonal Structure from a Simple Neural Network - Colab](#)

Jason Zhang



Source: [How to use the Circle of Fifths to write songs - Neel Modi - Words. Music. Images.](#)

Introduction

Music perception is not just predictive, but also inherently structured: listeners would quickly establish a “tone center” when hearing the music, group certain chords as their “favorites”, and experience emotion fluctuations through key changes or chord progressions. These traits emerge even without special musical training, which suggests that tonal knowledge reflects internalized structure rather than externally implemented-rules and or formal theoretical reasoning.

This leads to a central question in music cognition, that if such tonal structure can be represented symbolically using rule-based reasoning, or if it can emerge from learned associations.

Connectionist models have vouched for the latter view, and propose that key perception and tonal stability emerge as patterns of activation shaped by exposure to musical regularities rather than explicit theoretical knowledge (Bharucha & Todd, 1989).

Inspired by the connectionist models and my own passion for music, I created this project in an attempt to recreate similar results from those connectionist models of tonal perception by building a small, simple neural network that learns associations between chords and musical keys. Instead of aiming for classification accuracy like I did earlier in the semester, this final

project emphasizes on examining the structure of the network's output and internal representations, and interpreting them to how humans understand music in relation to cognition.

My plan for the model would include three criteria: 1) Showing graded key induction as a function of harmonic distance. 2) Showing dynamic key tracking across a chord sequence containing modulation. 3) Showing the geometry of hidden-layer representations. I conducted analyses to test whether familiar tonal structures can in fact emerge from simple learned associations.

Methods

Representations:

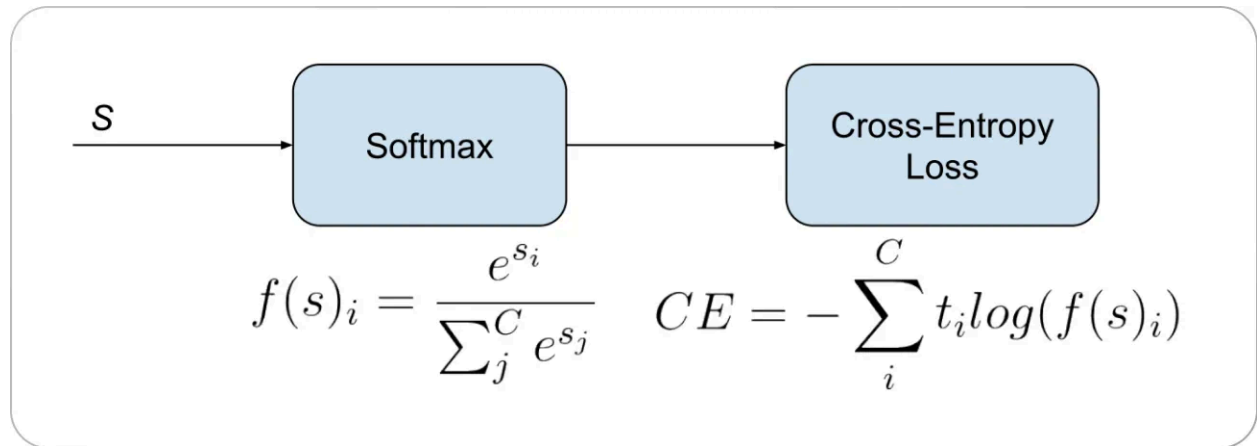
Chords were represented as one-hot vectors corresponding to the 12 major and 12 minor triads, 24 in triads in total. Musical keys were represented similarly with separate units for 12 major and 12 minor keys. Importantly, these one-hot representations contain no inherent musical similarity: each chord is treated as its own unique category with no shared structure. As a result, any harmonic relationships can only be learned from the statistical structure of the training data.

Training data:

Synthetic data (not real data) were generated for this training purpose to approximate exposure to tonal music. For each musical key, diatonic triads were sampled with weighted probabilities. This makes tonic and dominant chords appear more frequently than other chords with different

scales. This creates an unbalanced distribution of chords in tonal music, which reflects the statistical regularities and serves as representative for long-term musical exposure.

Model architecture:



A small feed-forward neural network with a single hidden layer. Chord inputs were mapped to key outputs via a 48-unit hidden layer with ReLU activation, and ended with a softmax output layer. The network was trained using cross-entropy loss and the Adam optimizer.

Evaluation procedures:

I evaluated the model performance in a more “cognitive” way than focusing on pure accuracy like what we did in the Music Classifier Model earlier in the semester. The results were interpreted in 3 directions:

- Key induction gradients measured as output probabilities across keys ordered by circle-of-fifth distance.

- Key tracking over time using a leaky integration of hidden activations applied at inference time to simulate contextual accumulation
- PCA and t-SNE applied to hidden layer activations to visualize representational structure

Results

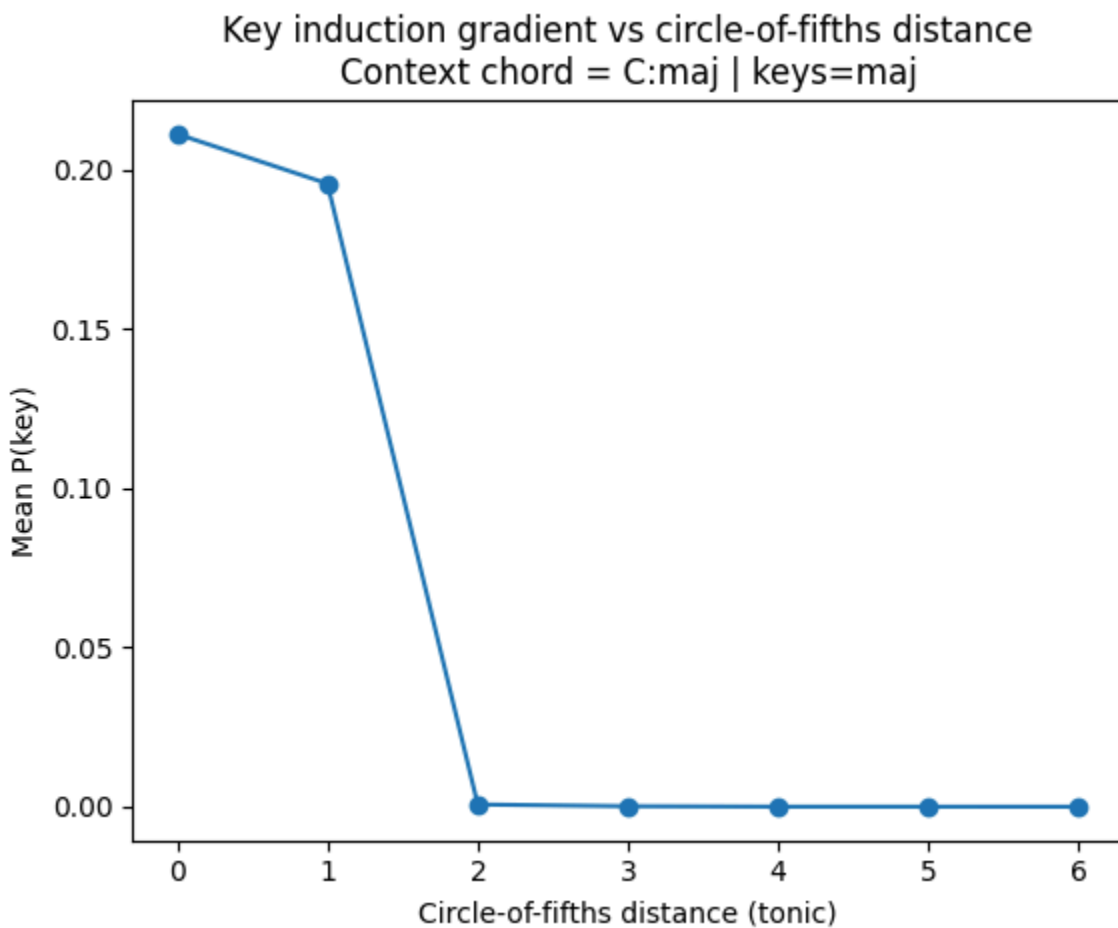


Figure 1. C major context chord.

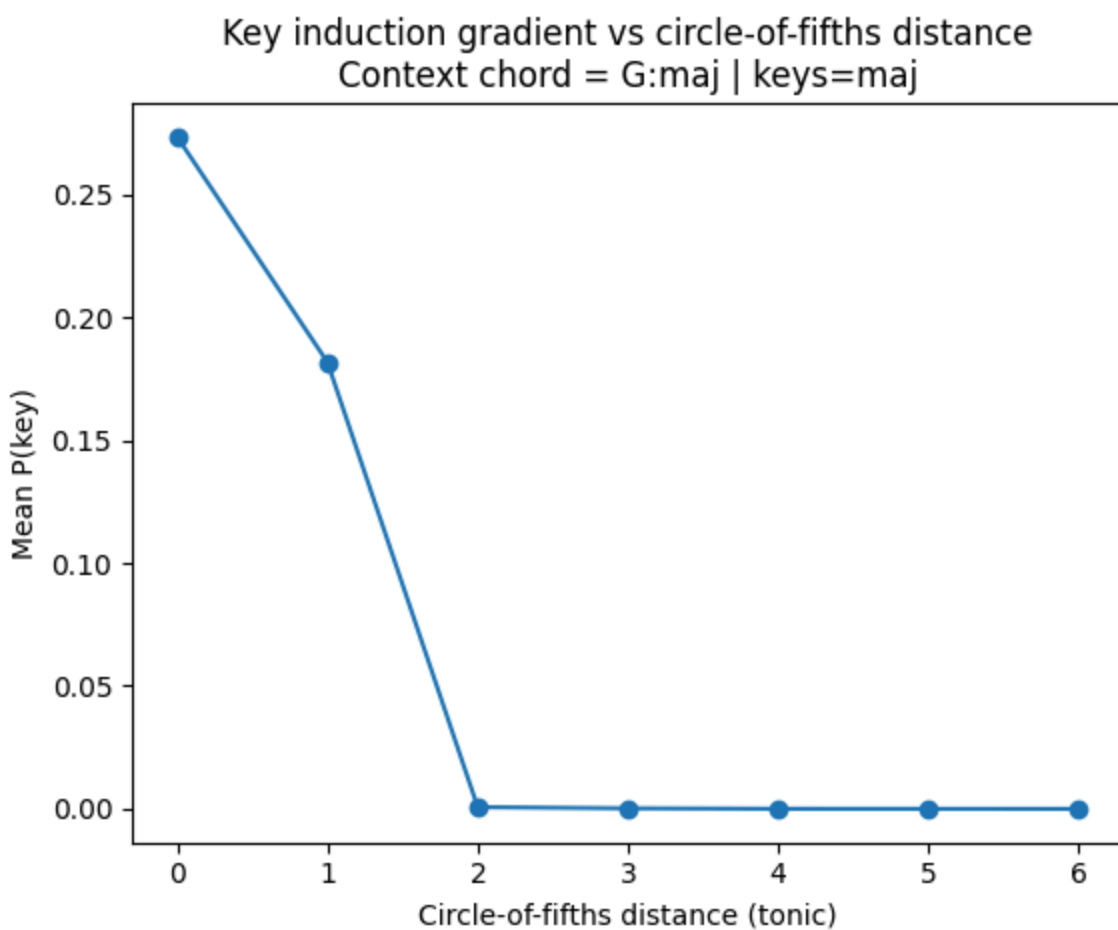


Figure 2. G major context chord

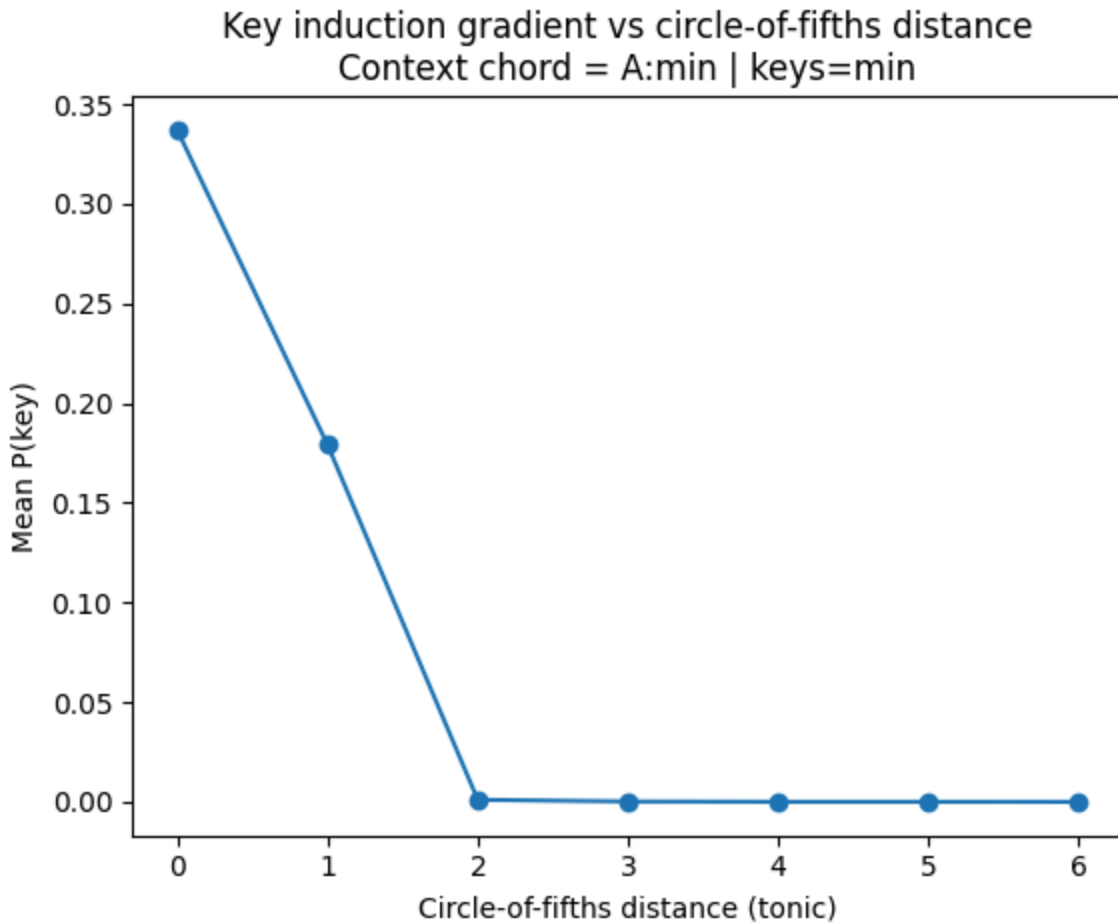


Figure 3. A minor context chord.

About the plots:

Figure 1, 2, and 3 all show key induction gradients for the three context chords here, including C major, G major and A minor. Across all three context chords, the highest output probability is assigned to the corresponding tonic key. As distance on the circle of fifths increases, the probabilities drop sharply (Distance $\Rightarrow$ 2).

Interpretation:

This pattern illustrates a similar pattern like human tonal judgements: close-relative keys are determined more plausible than distant ones. Interestingly, this structure was not explicitly built in the codes but rather emerged from the network's self-behavior, which indicates learning associations. This supports the idea that tonal hierarchies can emerge from statistical exposure.

Key tracking and modulation:

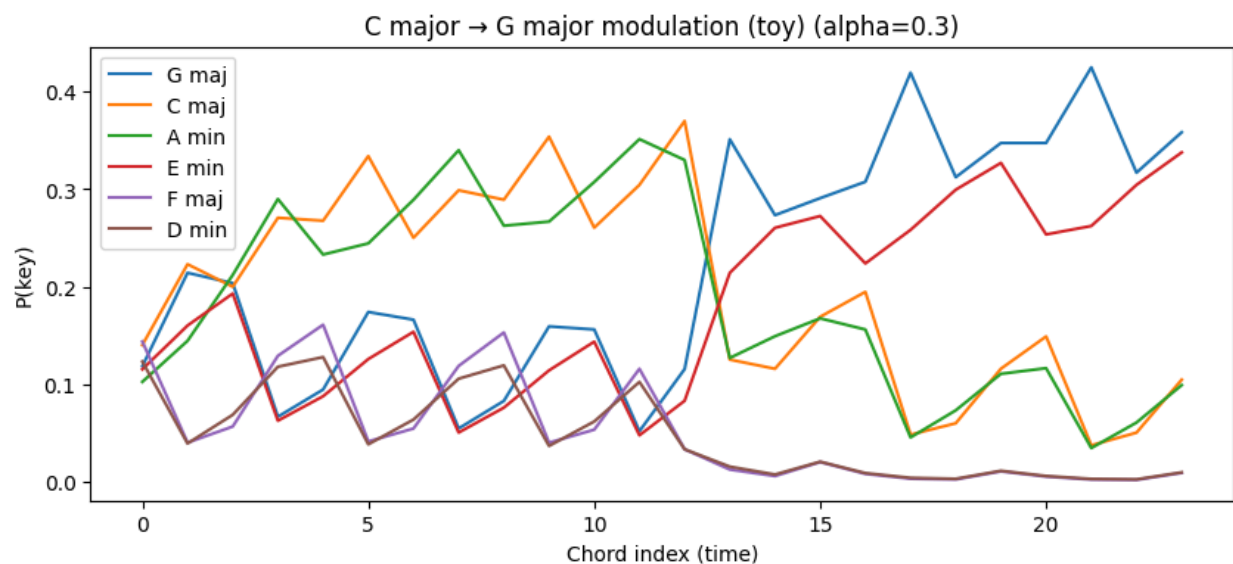


Figure 4. Key probability trajectories over time for modulating C major to G toy chord progression. Early in the chord index, C major shows the highest activation, as well moderate activation from related keys such as A minor chord.

Chord sequence:

C:maj | G:maj | A:min | F:maj | C:maj | G:maj | A:min | F:maj | C:maj | G:maj | A:min | F:maj |
 G:maj | D:maj | E:min | C:maj | G:maj | D:maj | E:min | C:maj | G:maj | D:maj | E:min | C:maj

We can see that as the C-centered chords shift toward G-centered chords, the probability gradually shifts toward G major and E min and the probability for C major and A minor gradually drops. This transition reflects contextual accumulation and resembles human listeners' experience: a gradual rethink and reconstruction, rather than an instant switch.

Hidden layer representations:

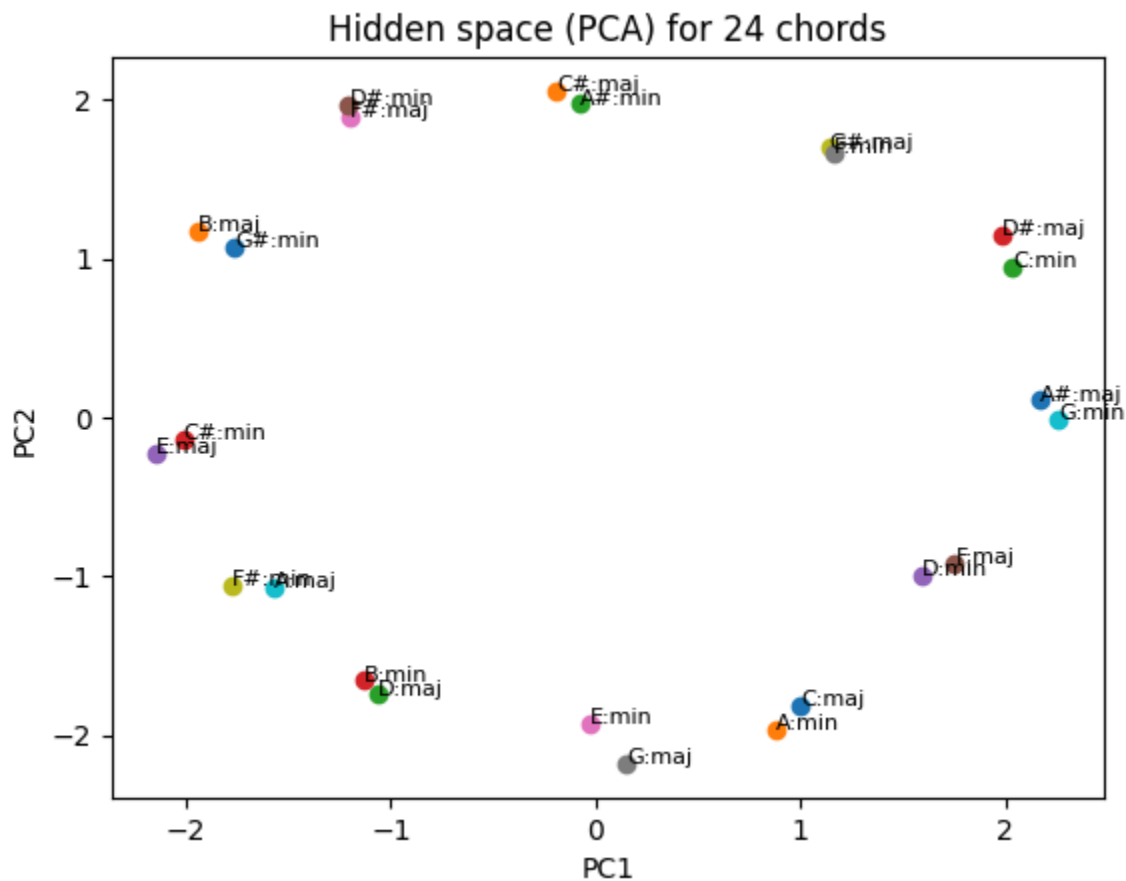


Figure 5. Hidden space PCA projection

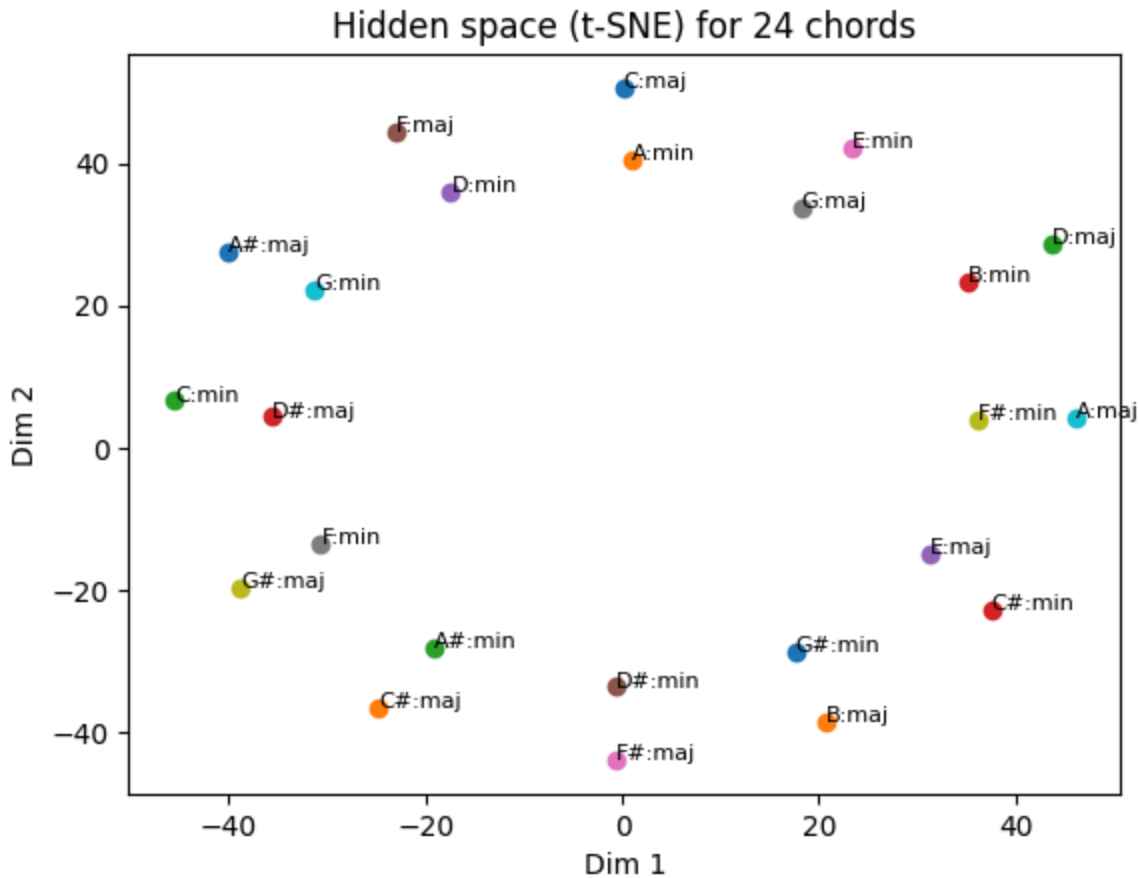


Figure 6. Hidden space t-SNE projection

Interpretation:

Figure 5 (PCA) and Figure 6 (t-SNE) are projections of hidden-layer activations by individual chord inputs. As we can see, even though the model was not given pre-written rules about the chord arrangement, the hidden space was able to self-organize a clear structure rather than completely randomized dispersion.

Based on the pattern from both figures, we can say that chords with related harmonic functions tend to form small clusters and overall an approximately circular pattern in positions, just like the

real concept of circle-of-fifths in music. This suggests that the network encodes harmonic similarity within its internal representation model. This supports the connectionist claim that musical categories may be represented as positions in a continuous space, not as discrete symbolic labels.

Discussion

Although the model is fairly straight-forward, it does successfully reproduce some “key” features of human tonal perception, including graded key induction, gradual key tracking across modulation, and clear-structured internal representations. This supports the idea that tonal cognition can emerge from learned associations rather than just explicit symbolic rules.

Value accuracy:

Epoch 01 | train loss 2.8535 | val loss 2.2754 | val acc 0.288

Epoch 02 | train loss 1.9776 | val loss 1.8405 | val acc 0.289

Epoch 03 | train loss 1.7998 | val loss 1.7670 | val acc 0.289

Epoch 04 | train loss 1.7563 | val loss 1.7425 | val acc 0.285

Epoch 05 | train loss 1.7401 | val loss 1.7327 | val acc 0.286

Epoch 06 | train loss 1.7324 | val loss 1.7269 | val acc 0.283

Epoch 07 | train loss 1.7281 | val loss 1.7238 | val acc 0.290

Epoch 08 | train loss 1.7252 | val loss 1.7193 | val acc 0.285

Epoch 09 | train loss 1.7233 | val loss 1.7187 | val acc 0.283

Epoch 10 | train loss 1.7222 | val loss 1.7172 | val acc 0.286

Epoch 11 | train loss 1.7213 | val loss 1.7162 | val acc 0.286

Epoch 12 | train loss 1.7203 | val loss 1.7167 | val acc 0.287

When we look at the classification accuracy (approximately 29%), it is fully expected due to the inherent ambiguity of mapping single chords to 24 possible keys, and requires extended training to become more deterministic. Because of that, it is theoretically impossible to have perfect accuracy in such a setting. Instead, the observed “low” accuracy reflects a more appropriate and cognitive representation of human tonal ambiguity, as multiple keys remain plausible given limited context.

Limitations:

Due to the simplicity, there are several limits. First, these data are not real-world data, and musical factors such as temporal and rhythmic structure are not considered in this model. Additionally, this network is a simple feed-forward network, which means the current architecture lacks learned temporal dynamics from a recurrent-network design.

Future directions:

For any further work, I may plan to extend the model by adding a tone-set-to- chord stage, conducting training based on real chord sequences from simple real-world melodies (For example, Twinkle Twinkle Little Star), or comparing human probe-tone behavior to the model outputs. These directions will help us further test about tonal structure’s emergence from learned statistical regularities.

Reference:

Bharucha, J. J., & Todd, P. M. (1989). Modeling the Perception of Tonal Structure with Neural Nets. *Computer Music Journal*, 13(4), 44–53. <https://doi.org/10.2307/3679552>